

Real Analysis Requests

What the majority of real bacteria/virus questions actually look like

54.6% of 5,250 real Biostars questions (bacteria/virus/pathogen-related) are "analysis request" shaped — data described, a goal stated. Here are two of them, in full, unedited.

Source: Luna, A. (2023). *BioStars Posts API Output*. Zenodo.

<https://doi.org/10.5281/zenodo.7813785> — CC BY 4.0

Example 1 — Bacterial SNP Detection (1/2)

Biostars #96189 · March 26, 2014 · biostars.org/p/96189

Hi all,

I am working on 10 bacterial genomes (1 reference and 9 mutant) sequenced by Illumina technology. My main aim is to find SNPs that are common in 9 genomes but absent in reference genomes. In last, I would like to do the automatic annotation of those SNPs. Until now, I have done the following steps and wondering if I am on the right path.

First: Extracted the common SNPs in 9 mutant genomes

```
vcf-isec -n +9 -f 1.vcf.gz 2.vcf.gz 3.vcf.gz 4.vcf.gz 5.vcf.gz 6.vcf.gz 7.vcf.gz 8.vcf.gz 9.vcf.gz | bgzip -c > isec1.vcf.gz
```

Second: tab index

```
tabix -p vcf isec1.vcf.gz
```

(continued on next slide — full post, unedited)

Example 1 — Bacterial SNP Detection (2/2)

Biostars #96189, continued

Third: Extracted SNPs present in isec1.vcf.gz but absent in reference strain

```
vcf-isec -c -f isec1.vcf.gz reference.vcf.gz > isec2.vcf
```

Four: Automatic annotation of isec2.vcf. Used snpEFF:

```
java -jar snpEff.jar eff -no-downstream -no-upstream -no-utr -no-intergenic -v database isec2.vcf
```

Most of the SNPs were observed in intergenic region. Should I include these intergenic SNPs or not? Any other suggestions of selecting SNPs.

Regards, Nitin

Example 2 — Viral Nanopore Read Downsampling

Biostars #398159 · September 11, 2019 · biostars.org/p/398159

Tools for downsampling/resampling/subsampling Oxford Nanopore reads

I have got ~40000 MinION reads derived from a virus, whose read length ranges from ~100 to ~34.9k. I would like to re-sample these reads to evaluate the minimum read number enough to detect this virus. As Oxford nanopore reads have no identical read length, I have to take the varying read length into consideration when re-sampling the reads. Are there open source tools available for this task?

Notice what both examples share: a concrete data description (organism, platform, read count/type) and a stated goal — even though neither writer was told to phrase it that way. That's the shape 54.6% of real questions take, and it's exactly the shape `route.py` and the rest of this pipeline are built to consume.